SMART LOCATION - BASED HOSPITAL FINDER SYSTEM

PROJECT REPORT

Submitted by

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(An ISO 9001:2015 certified Institution)

(Accredited by NAAC with A++)

Pollachi-642 001

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CERTIFICATE – I

This is to certify that the project work entitled **"SMART LOCATION - BASED HOSPITAL FINDER SYSTEM"** is a bonafide record of the work done by P.SANGEETHA (23-CE-049) doing III-B.Sc. Computer Science (SF) submitted to Nallamuthu Gounder Mahalingam College, affiliated to Bharathiar University in partial fulfilment of the requirements for the award of the degree of B.Sc. Computer Science (SF) under my supervision.

GUIDE **HEAD OF THE DEPARTMENT**

This is to certify that this is a bonafide record of the work submitted by

P. SANGEETHA

(23-CE-049)

For External Viva-voce examinations held on …….

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This is to certify that the project work **entitled "SMART LOCATION - BASED HOSPITAL FINDER SYSTEM"** is the original work done by P. SANGEETHA (23-CE-049) under my guidance**.**

Date :

Place : **Signature of the Guide**

Ms. G. ANGAYARKANNI, M.C.A.,M.Phil.,B.Ed.,M.E.,(Ph.D.,)

CERTIFICATE-III

This is to certify that the project work entitled **"SMART LOCATION - BASED HOSPITAL FINDER SYSTEM"** is the original work done by me under the guidance of **(Ms. G. Angayarkanni, M.C.A., M.Phil., B.Ed., M.E., (Ph.D).,)**

Date :

Place :

P. SANGEETHA

(23-CE-049)

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I am grateful to express my thanks to my parents, whose gracious encouragement allowed me to work with a clean mind and heart.

I also thank my friends, who have helped me in completing the project successfully.

SYNOPSIS

The Smart Location-Based Hospital Finder System is a web-based application developed using PHP and MySQL to help users quickly find nearby hospitals based on their location and requirements. The main aim of this project is to provide an easy and efficient way for patients to search for hospitals without wasting time, especially during emergency situations.

In this system, the patient can register and log in, then enter their city or location and select the type of hospital such as general, government, private, or emergency. The system will display the list of hospitals available in that location along with details such as hospital name, address, and contact number.

The project also includes Admin and Doctor modules. The Admin can manage hospital information, view patient details, and maintain the database. The Doctor can log in to the system and view patient details and reports. All the information is stored securely in the MySQL database.

This system reduces the difficulty of searching for hospitals manually and provides quick access to healthcare services. The Smart Location-Based Hospital Finder System is useful for patients, doctors, and administrators to manage hospital information efficiently and improve healthcare accessibility.

1. INTRODUCTION

1.1 Overview of the project

The project entitled **"Smart Location-Based Hospital Finder System"** is a web-based application developed using PHP with MySQL as the backend database. The main purpose of this system is to help users easily find nearby hospitals and doctors using a smart location feature. The application can be accessed by users at any time and from any place through the internet, which makes it very useful during emergency situations.

In many cases, people require immediate medical assistance but face difficulty in finding a nearby hospital or available doctor. This system helps solve that problem by allowing users to search for hospitals and doctors based on their location. It provides details such as hospital name, doctor specialization, contact information, and available appointment times. With this information, patients can quickly identify the nearest medical facility and receive medical care without delay.

The system also provides an online health guidance feature through a baby care And symptom-based consultation module. In this module, users can enter their symptoms And health issues. The system contains a database that stores various diseases and their Related symptoms. When the user submits symptoms, the system processes the information And predicts possible illnesses that may be associated with those symptoms. Based on this analysis, the system provides basic guidance and suggests consulting a doctor for proper treatment.

In the doctor module, doctors can log in to the system and view patient details, appointment schedules, and patient reports. Doctors can also view the symptoms searched by patients and provide appropriate consultation based on the predicted results. This helps doctors understand the patient's health condition before the consultation process.

The admin module manages the overall system. The admin can add new disease information by entering the disease name, type, and symptoms into the database. The system uses a data mining technique to analyze the relationship between symptoms and diseases. The admin can also view, update, and manage doctor information, patient records, and disease details stored in the system.

Another important feature of the system is doctor availability and appointment booking. Doctors can set their consultation time slots in the system. When a patient tries to book an appointment at a particular time, such as 10:00 AM, the system checks the database to see if another appointment already exists for that doctor at the same time. If the time slot is already booked, the system displays a message asking the user to choose another time. If the slot is available, the appointment will be successfully booked.

The system also includes a feedback facility where patients can share their experience about the doctor and hospital services after consultation. This feedback helps improve healthcare service quality and assists other users in selecting suitable doctors.

Overall, the Smart Location-Based Hospital Finder System provides an efficient and user-friendly platform for finding hospitals, booking appointments, receiving health guidance, and improving communication between patients and healthcare providers.

1.2 Description of the Module

Modules of the project as follows

- Patient Registration Module
- Patient Login Module
- My Details Module
- Disease Prediction Module
- Search Doctor Module
- Doctor Login Module
- Patient Details Module
- Doctor Availability Time Module
- Appointment Booking Module
- Appointment Time Checking Module
- Feedback Module
- Admin Module

Admin Module

The Admin Module is responsible for managing the overall system. The administrator logs into the system using a secure username and password. After login, the admin can add, update, and delete doctor details, manage disease and symptom information, and view patient records. The admin also monitors system activities and manages feedback given by patients. This module ensures the smooth functioning and maintenance of the hospital finder system.

Patient Registration Module

The Patient Registration Module allows new users to create an account in the system. The patient enters personal information such as name, address, gender, email, and password. After registration, the system stores the patient details in the database and provides login credentials. This module ensures that every patient has a unique identity in the system.

Patient Login Module

The Patient Login Module allows registered patients to access the system using their user ID and password. After successful login, the patient can use various features such as searching for doctors, checking disease predictions, booking appointments, and viewing personal details. This module ensures secure access to the application.

Disease Prediction Module

The Disease Prediction Module helps patients identify possible illnesses based on their symptoms. The patient enters symptoms related to their health condition. The system compares these symptoms with the diseases stored in the database and predicts the possible illness using a data mining algorithm. Based on the prediction, the system may suggest suitable doctors for consultation.

Search Doctor Module

The Search Doctor Module helps patients find doctors easily based on different criteria such as doctor name, specialization, or location. The system displays doctor details including specialization, contact number, and consultation fees. This feature helps patients quickly locate the right doctor for their health problem.

Doctor Module

The Doctor Module allows doctors to log into the system and view patient information and appointment details. Doctors can view the symptoms entered by patients and the disease predicted by the system. This helps doctors understand the patient's condition before providing consultation.

Doctor Availability Time Module

The Doctor Availability Time Module allows doctors to specify their available consultation time in the system. Doctors can set their working hours and available time slots for appointments. Patients can view these available time slots before booking an appointment. This helps in proper schedule management and avoids time conflicts.

Appointment Booking Module

The Appointment Booking Module allows patients to book appointments with doctors. The patient selects the doctor, appointment date, and preferred time slot. The system records the appointment details in the database and confirms the booking if the selected time slot is available.

1.3 System Specification

1.3.1 HARDWARE SPECIFICATION

System : Pentium IV 2.4 GHz.

Hard Disk : 40 GB.

Floppy Drive : 1.44 Mb.

Monitor : 14' Colour Monitor.

Mouse : Optical Mouse.

Ram : 512 Mb.

Keyboard : 101 Keyboard.

1.3.2 SOFTWARE SPECIFICATION

User Interface Design : HTML, AJAX,JQUERY,JAVASCRIPT

Web Browser : Mozilla, Google Chrome, IE8,OPERA

Software : XAMPP Server

1.4 Software Features

Front end : PHP

Backend : My SQL

1.4.1 Front End

PHP is a powerful server-side scripting language for creating dynamic and interactive websites. PHP widely used; free and efficient alternative to competitors such as Microsoft's ASP.PHP is perfectly suited for Web development and can be embedded directly into the HTML code. The PHP syntax is similar to pearl and C.

PHP is open source that it is readily available and absolutely free. Stability, flexibility and speed are chief qualities that attract to choose PHP.PHP have multiple extensions and is extremely scalable.

Server-side scripting:

This server-side scripting is the most traditional and main target field for PHP. Programmer needs three things to make this work. Programmer need to run the web server, with a connected PHP installation. Programmer can access the PHP program output with a web browser, viewing the PHO page through the server. All these can run on your home machine if programmers are just experimenting with PHP programming.

Command line scripting:

Programmer can make a PHP script to run it without any server or browser. Programmers only need the PHP parser to use it this way. This type of usage is ideal for scripts regularly executed using croon (on*nix or Linux) or Task Scheduler (on Windows).These scripts can also be used for simple text processing tasks.

Features of PHP

- PHP runs on different platforms (Windows, Linux, UNIX, etc.)
- PHP is compatible with almost all servers used today.
- PHP is free to download from the official PHP resource: www.php.net.

BACK END

MYSQL

MYSQL is an open-source relational database management systems (RDBMS), is developed, distributed and supported by MYSQL AB. MYSQL is a popular choice of database for use in web applications MYSQL can be scaled by deploying it on more powerful hardware, such as a multi-processor server with gigabytes of memory. MYSQL is easy to use, yet extremely powerful, secure, and scalable. And because of its small size and speed, it is the ideal database solution for Web sites.

MYSQL is a database management system

A database is a structured collection of data. It may be anything from a simple shopping list to a picture gallery or the vast amount of information in a corporation network. To add, access and process data stored in a computer database we need a database management system such as MYSQL server. Since computers are very good at handling large amount of data, database management system plays a central role in computing.

MYSQL is a relational database management system

A relational database stores separate data in separate tables rather than putting all the data in one big storeroom. This adds speed and flexibility. The SQL part of “MySQL” stands for “Structured Query Language”. SQL is the most common standardize language used to access database and is defined by the ANSI/ISO SQL standard. The SQL standard has been evolving since 1986 and several versions exist.

MYSQL software is open source

Open source means that it is possible for anyone to use modify the software. Anybody can download the MYSQL software uses the GPL(GNU General Public License), to define what we may and may not use do with the software.

MYSQL Server works in Client/ Server or embedded systems

The MYSQL database software is a client/server system that consists of a multi-threaded SQL server that supports different backend, several different client programs and libraries, administrative tools and a wide range of Application Programming Interface(APIs). A large amount of contributed MYSQL software is available: Modern day websites seem to be relying more and more on compel the Structured Query Language is a very popular database language, and its standardization makes it easy to store, update and access data. One of the most powerful SQL servers out there is called MYSQL and surprisingly enough, it's free.

Some of the features of MYSQL include: Handles large databases, in the area of 50,000,000+records. No memory leaks. Tested with a commercial memory leakage detector (purify). A privilege and password system which is very flexible and secure, and which allows host-based verification. Passwords are secure since all password traffic when connecting the server is encrypted.

Features of MYSQL

Client/server Architecture: MYSQL is a client/server system. There is a database server (MYSQL) and arbitrarily many clients (application programs), which communicate with the server. The clients can run on the same computer as the server or on another computer.

SQL Compatibility:

As before said SQL is a standardized language for querying and updating data and for the administration of a database. Through the configuration setting sol-mode we can make the MYSQL server behave for the most part compatibly with various database systems.

Stored procedures:

Stored procedures (SPs for short) are generally used to simplify sgteps such as inserting or deleting a data record.

Triggers:

Triggers are SQL commands that are automatically executed by the server in certain database operations INSERT, UPDATE, and DELETE, MYSQL has supported triggers.

Replication:

Replication allows the contents of a database to be copied (replicated) onto a number of computers to increase protection against system and to improve the speed of database queries.

Platform independence:

MYSQL can be executed under a number of operating systems. The most important are Apple Macintosh OS X, Linux, Microsoft Windows, and the Unix.

Speed:

MYSQL is considered a very fast database program.

2.SYSTEM STUDY

2.1 Existing System

In the existing system, most hospital activities are managed manually using paper records. Patient details, prescriptions, and laboratory appointments are written and maintained by hospital staff. This process requires more manpower and time to record, update, and maintain patient information. Because the records are maintained manually, there is a high possibility of errors, confusion, and loss of important data. From the patient's perspective, the process is inconvenient and time-consuming. Patients must visit different places such as testing laboratories and pharmacies to complete their medical procedures. They often have to wait in long queues for lab tests, medicines, and doctor consultations, which can be stressful especially during emergencies. From the administrator's side, managing hospital operations manually is very difficult. Maintaining patient records, prescriptions, and appointment details requires a lot of paperwork. It is also difficult to organize data, generate reports, and ensure proper management of hospital information. Therefore, the existing system is inefficient, time-consuming, and not suitable for managing modern healthcare services.

2.1.1 Drawback of Existing System

- The Necessary data are stored in different documents and transactions are noted in it.
- Many files and paper works are maintained by the existing system.
- Loss of the Messages, Data.
- The existing system has no security measures against logging in and no checks are made for authorized users.
- The system does not have any descriptive reports and thus did not help management in decision-making.
- Enormous amount of time is consumed.
- The data is not in secured format

2.2 Proposed System

The proposed system is designed to overcome the disadvantages of the existing manual system. In the current system, most hospital activities are handled manually, which leads to errors, confusion, and delays. To solve these problems, the proposed system introduces a computerized and automated solution for managing hospital information. This system reduces manual effort by storing patient, doctor, and disease details in a centralized database. It also minimizes time consumption by allowing users to easily search for doctors, check disease predictions based on symptoms, and book appointments online. Doctors can view patient details and appointment schedules, while the administrator can manage system data efficiently. Overall, the proposed system improves accuracy, saves time, and provides easy storage and retrieval of information.

2.2.1 Advantage of Proposed System

- The proposed system reduces maintenance cost and investment compared to the existing manual system.
- It prevents errors that usually occur due to manual data entry and paper-based record management.
- The system is user-friendly, making it easy for patients, doctors, and administrators to use the application without difficulty.
- The processing time is reduced because all data is stored and managed digitally.
- The system provides quick responses when searching for doctors, hospitals, or patient information.
- Retrieval of information becomes easy since all records are stored in a centralized database.
- The system can generate useful reports that help in better management and decision-making.
- The system is flexible and can be updated or modified according to future requirements.

SYSTEM DESIGN AND DEVELOPMENT

3.1 Input Design

Input design is the process of converting user-originated inputs to a computer- based format input data are collected and organized into group of similar data. Once identified, appropriate input media are selected for processing.

Login

The form contains the user name and password. By entering the user name and Password the user can login to the application.

Patient Login:

Patient Login to the system using his ID and Password.

Patient Registration:

Patient is a new user he will enter his personal details and he will user Id and password through which he can login to the system.

3.2 Database Design

A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and efficiently. The general objective to make information access easy, quick, inexpensive and flexible for user.

The Objective of Database Design:

- Control redundancy
- Easy of learning and use
- Data independency
- Accuracy and integrity
- Recovery from failure
- Privacy and security

3.2.1 Table Design

Table name : admin

Primary key : admin_id

Description : This table is to store the admin login details.

Field Name	Data Type	Width	Description
Admin_id	Varchar	10	The unique identification
Username	Varchar	25	User name or doctor name
Password	Varchar	25	Login Password
Updation_date	Date	-	Update date

Table name : doctors

Primary key : doctors id

Foreign key : specialization_id

Description : This table is to store the doctors login details

Field Name	Data Type	Width	Description
Doctor_id	Varchar	11	The unique identification
Specialization id	Varchar	35	specialization
DoctorName	Varchar	20	Doctor Name
Address	Varchar	140	Doctor address
DocFees	Int	05	Doctor Fees
Contactno	Int	10	Contact number
docEmail	Varchar	25	Doctor Email
Password	Varchar	10	Login password
Creation Date	Timestamp	-	Creation Date
Updation Date	Timestamp	-	Updating Date

Table name : specilization

Primary key : specilizationid

Description : This table is to store the doctor specialization Details

Field Name	Data Type	Width	Description
Specialization id	Varchar	10	Order identification
Specialization	Varchar	20	Select specialization identification
Creation Date	Date	-	Creation Date
Updation Date	Timestamp	-	Updating Date

Table name : user

Primary key : user id

Description : This table is to store the user login details.

Field Name	Data Type	Width	Description
Userid	Varchar	10	The unique identification
fullName	Varchar	25	Full Name
Address	Varchar	255	Address
City	Varchar	20	City
Disease	Varchar	30	Disease detailed
Gender	Varchar	05	Gender
Email	Varchar	25	Email
Password	Varchar	10	Password
RegDate	Date	-	Registration Date
UpdationDate	Timestamp	-	Updation Date

Table name : appointment

Primary key : appointment id

Foreign key : UserId,Specializationid, DoctorId

Description : This table is to store the appointment Details

Field Name	Data Type	Width	Description
Appointment	Varchar	10	The unique identification
Specializationid	Varchar	20	Doctor specialization id
DoctorId	Varchar	11	Doctor identification
UserId	Varchar	11	User Id
Consultancy Fees	Int	05	Consultancy Fees
AppointmentDate	Date	-	Appointment Date
AppointmentTime	Time	-	Appointment Time
PostingDate	Date	-	Posting Date
UserStatus	Varchar	20	User status
DoctorStatus	Varchar	20	Doctor Status
UpdationDate	timestamp	-	Updating Date

3.2.3 E-R Diagram

The relation upon the system is structure through a conceptual ER-Diagram, which not only specifics the existential entities but also the standard relations through which the system exists and the cardinalities that are necessary for the system state to continue.

The entity Relationship Diagram (ERD) depicts the relationship between the data objects. The ERD is the notation that is used to conduct the date modeling activity the attributes of each data object noted is the ERD can be described resign a data object descriptions.

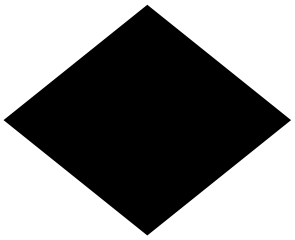
The set of primary components that are identified by the ERD are

- Data object
- Relationships
- Attributes
- Various types of indicators.

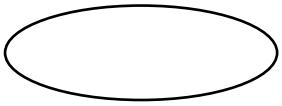
The primary purpose of the ERD is to represent data objects and their relationships.

ER-DIAGRM SYMBOL

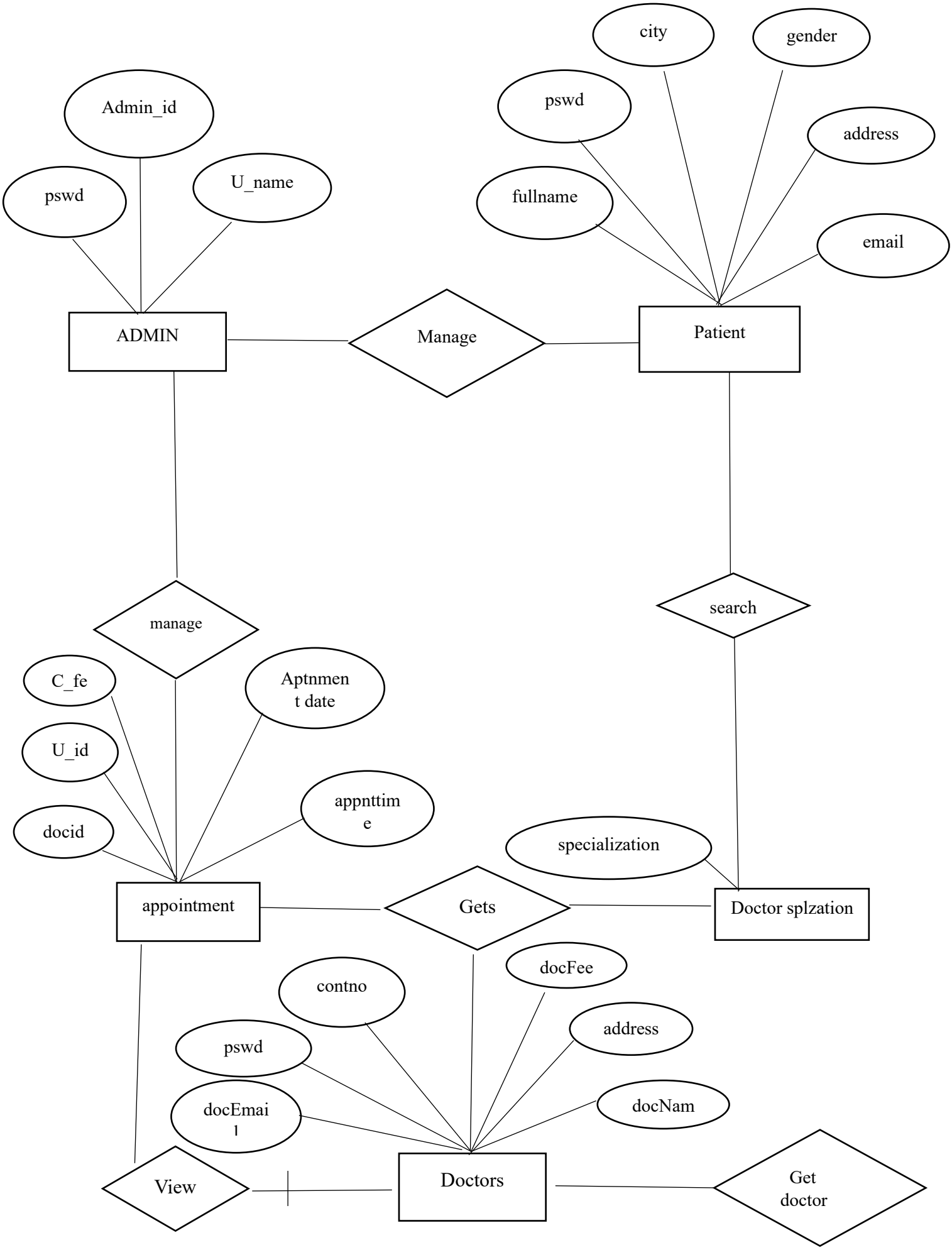
- Entity



- Relationship



- Attributes


Admin_id
pswd
U_name
ADMIN
Manage
Patient
city
gender
pswd
address
fullname
email
manage
search
C_fe
Aptnmen
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l
docNam
View
Doctors
Get
doctor

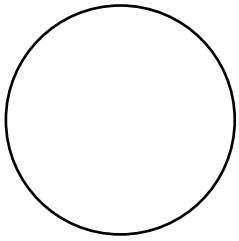
3.2.3 Dataflow Diagram

System design is the process of planning a new system to complement or altogether replace the old system. The purpose of the design phase is the first step in moving from the problem domain to the solution domain. The design of the system is the critical aspect that affects the quality of the software. System design is also called top-level design.

Data Flow Diagram Symbols

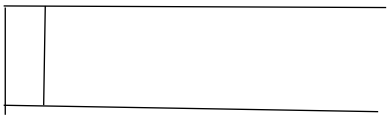
- Entity

- Dataflow

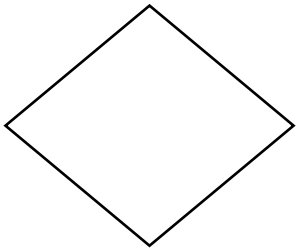


-

- Process

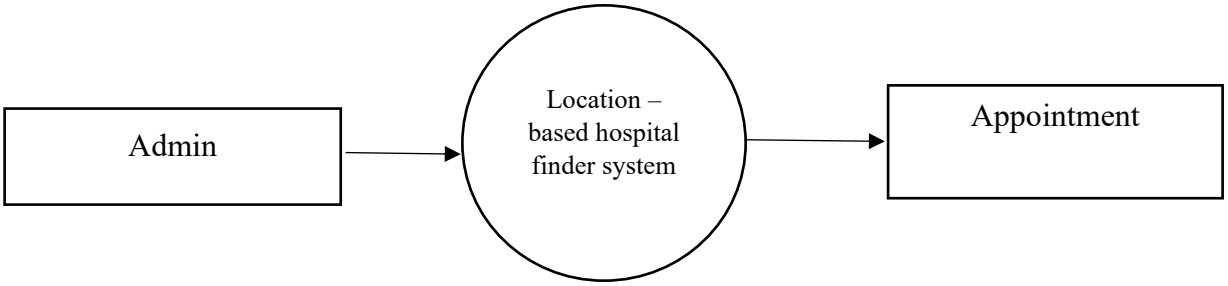


-Storage

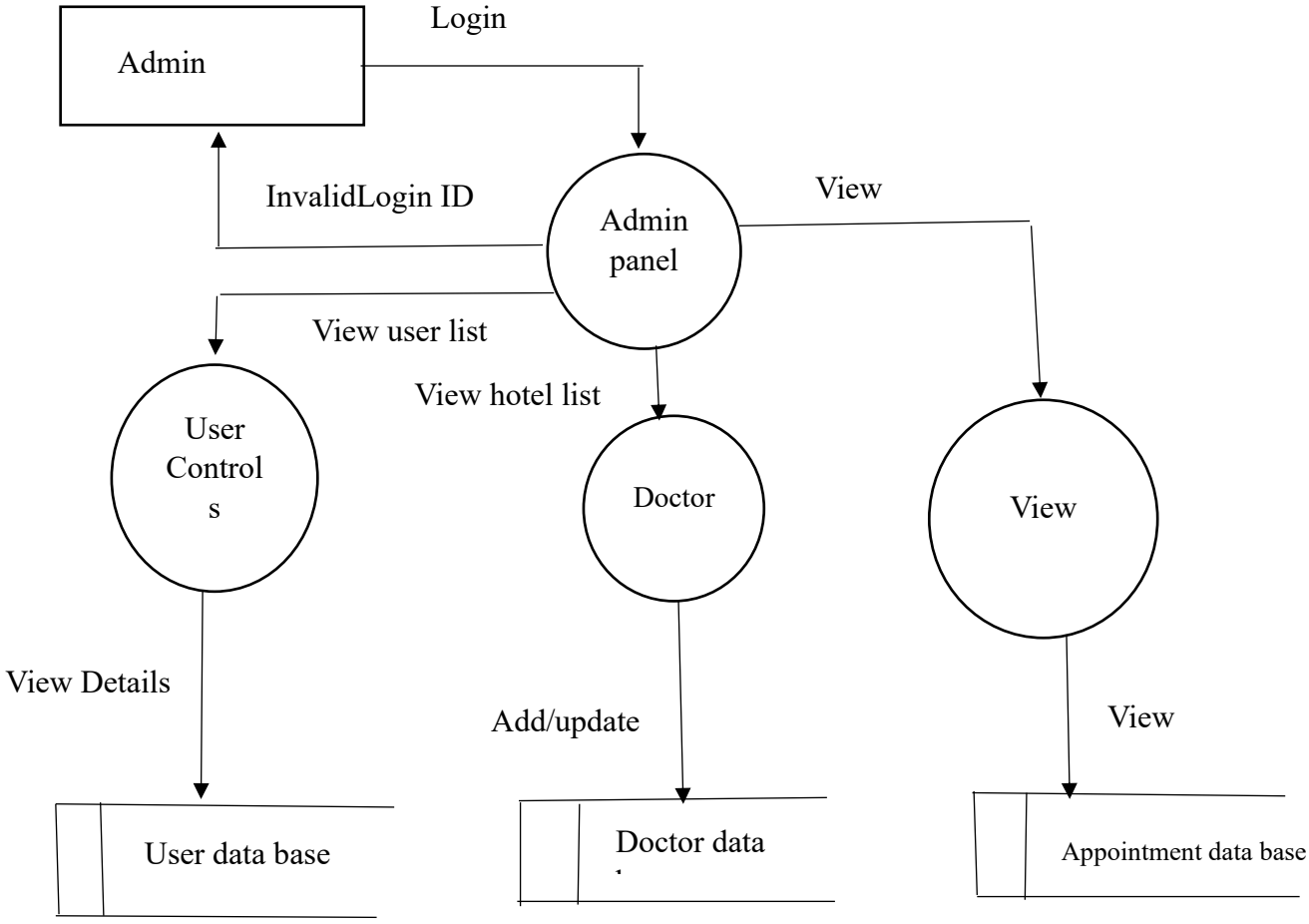


-Decision

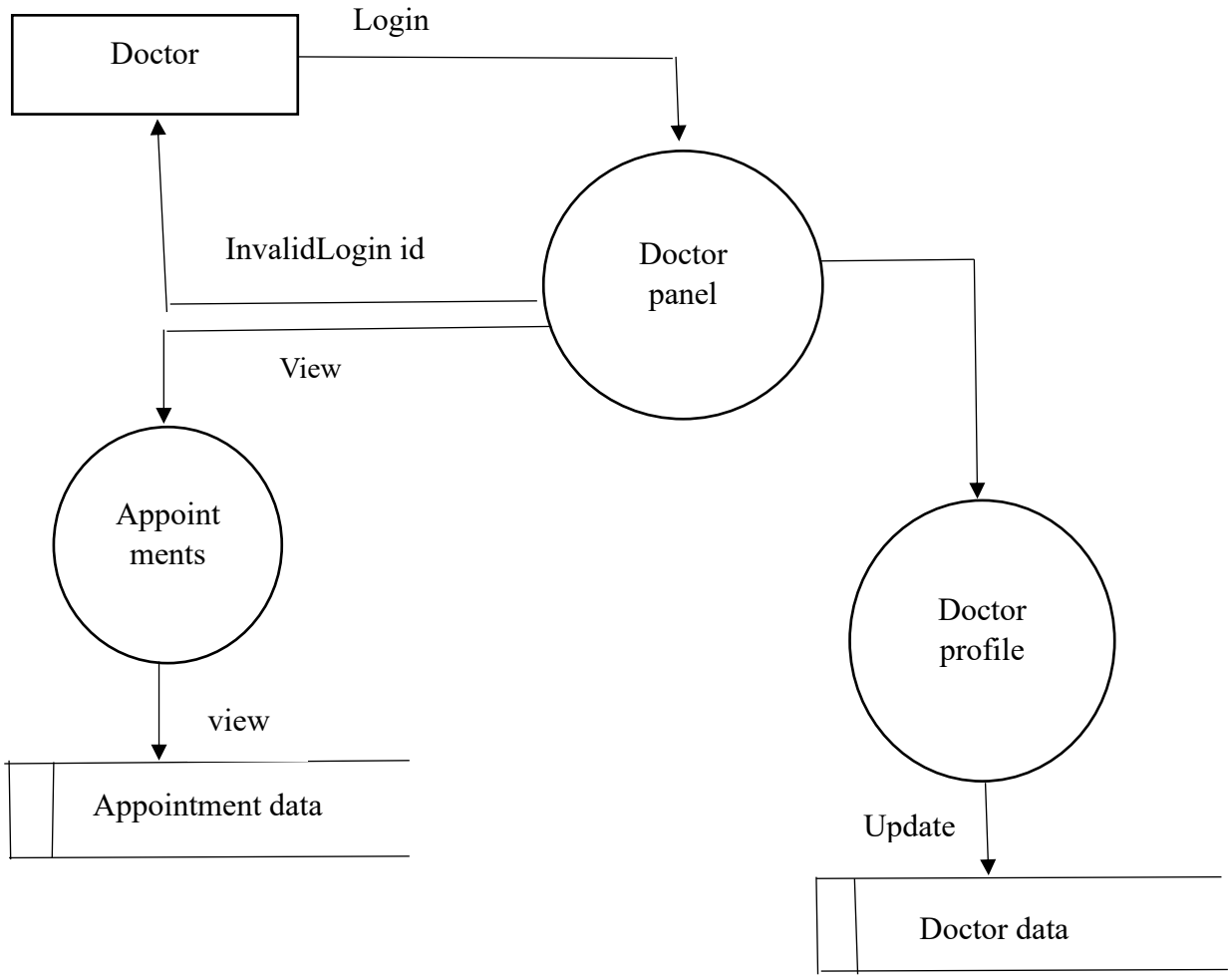
LEVEL 0: CONTEXT LEVEL DIAGRAM:



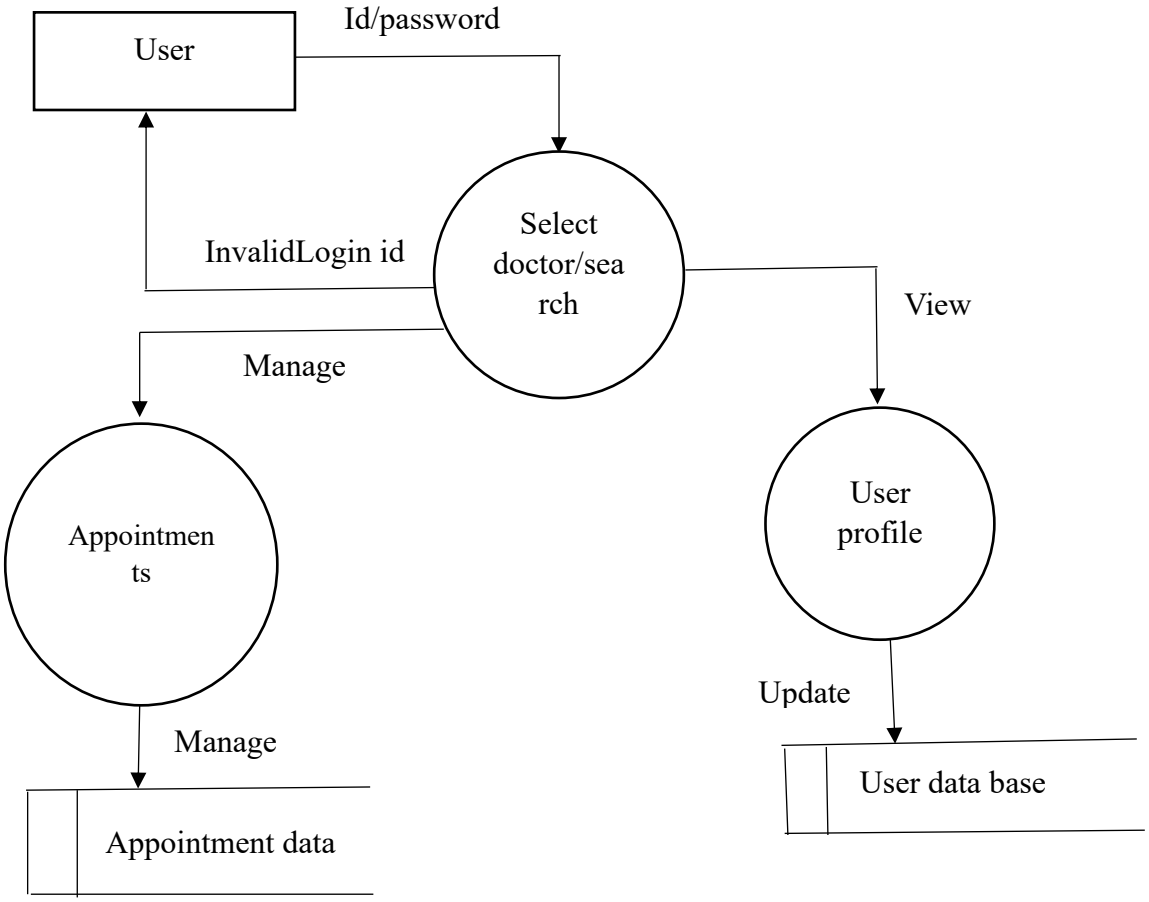
LEVEL 1 DFD OF ADMIN:



LEVEL 2 DFD OF DOCTOR:



LEVEL 3 DFD OF USER:



3.3 Code Design

The code design is to shows the all of project process. Code design contains the table's primary key details only. In case the primary key is email then the code design follows the structure of email such as abc@domain.com. Any strings and integers include the email id, the integers and strings are explain separately.

1.Registration Email

xxxxxxxxxxxxxxx 999

Email ID

String indicates that Email

Example:

Registration Email

abc 123

Email ID

String indicates that Email

2.Package **Package id**

xxxxxxxxxxxx **999**

Package ID

String indicates that Package name

Example :

Insert **Package**

abc **123**

Package ID

String indicates that Package

3.4 Output Design

1. Login Output

Description

- The login output screen allows the user to enter their User ID and Password to access the system. The system verifies the login credentials from the database. If the username and password are correct, the user will successfully log in to the system.
- If the details are incorrect, the system displays the message "Invalid Username or Password."

This output ensures secure access to the application.

2. Search Doctor Output

Description

The search doctor output screen allows patients to find doctors easily. The patient can search by entering details such as:

- Doctor Name
- Specialization
- Location

After searching, the system displays the list of doctors with details such as:

- Doctor Name
- Specialization
- Address
- Contact Number

This helps the patient quickly locate the required doctor.

3. Notification Output

Description

The notification output screen is available in the Doctor Module. In this screen, the doctor can view notifications related to the system.

The notification may include:

- Number of patients accessing the system
- Predicted diseases based on symptoms
- Patient activity in the system

This helps doctors understand the patient's health concerns and system usage.

4.My Details Output

Description

The My Details output screen allows the patient to view their personal information stored in the system.

The details displayed include:

- Patient Name
- Address
- City
- Gender
- Email ID

This screen allows patients to verify their information and update their profile if required.

4. SYSTEM TESTING AND IMPLEMENTATION

4.1 System Testing

The quality of the application can and normally does vary widely from system to system but some of the common quality attributes include reliability, stability, portability,maintainability and usability.

- Testing is a process of executing a program with the intent of finding an error.
- A good test case is one that has a high probability of finding an as yet

undiscovered error.

- A successful test is one that uncovers an as yet undiscovered error.

Testing should systematically uncover different classes of errors in a minimum amount of time and with a minimum amount of effort. A secondary benefit of testing is that it demonstrates that the software appears to be working as stated in the specifications.

The data collected through testing can also provide an indication of the software's reliability and quality. But, testing cannot show the absence of defect -- it can only show that software defects are present.

Types of System Testing for Smart Location-Based Hospital Finder System

1. Unit Testing

TC_ID	Module	Description	Expected output	Actual output	Status(pass/Fail)
Unit001	Login	Incorrect Username	Application will be login	Invalid user name &password	Pass
Unit002	Login	Correct Username & password	Application will be login	Login to the software	Pass

2.Integration Testing

Integration Testing is performed to check whether different modules of the system work together correctly. In the Smart Location-Based Hospital Finder System, this testing ensures that modules such as patient, doctor, admin, and database are properly connected and data flows correctly between them.

Integration Testing Table

IT_ID	Module	Description	Excepted Output	Status
IT001	Login+database	Verify login with database	User login successful	Pass
IT002	Patient+Doctor	Patient searches doctor	Doctor details displayed	Pass
IT003	Admin+database	Admin adds doctor/disease	Data stored in database	Pass
IT004	Doctor+patient	Doctor views patient details	Patient information displayed	Pass

3.Acceptance Testing

Acceptance Testing is performed to check whether the system meets the user requirements and works correctly for end users. In the Smart Location-Based Hospital Finder System, this testing ensures that major functions such as patient registration, login, doctor search, and admin management work properly. If the system performs all required operations successfully, it is considered to have passed the acceptance testing.

Acceptance Testing Table

AT_ID	Module	Description	Expected Output	Status
AT001	Patient registration	Enter patient details	Patient registered successfully	Pass
AT002	Patient login	Enter valid username and password	Login successful	Pass
AT003	Search doctor	Search doctor by name/specialization	Doctor details displayed	Pass
AT004	Admin login	Enter admin credentials	Admin panel opened	Pass

4.Validation Testing

In software testing verification and validation is the process of checking that a software system meets specifications and that it fulfills its intended purpose. It may also be referred to as software quality control.

Validation Testing Table

VT_ID	Module	Description	Expected Output	Status
VT001	Login	Enter valid username and password	Login successfully	Pass
VT002	Registration	Enter patient details	Patient registered successfully	Pass
VT003	Search	Enter doctor name/specialization	Doctor details displayed	Pass

4.2 Test Case

4.2.1 User Authentication

Test case ID	Test Scenario	Steps to execute	Expected result	Actual result	Status	Remark
UA-001	User registration	1.Navigate to registration page 2. enter valid details 3.Click register	Account successfully created	Account successfully created	Pass	Working properly
UA-002	Invalid registration	1. Navigate to registration page 2. Enter invalid/missing details 3. Click Register	Display validation error	Validation error displayed	Pass	Error handled
UA-003	User Login	1. Navigate to login page 2. Enter valid credentials 3. Click Login	User is logged in	User logged in successfully	Pass	Working properly
UA-004	Incorrect Password	1. Navigate to login page 2. Enter incorrect password 3. Click Login	Display error message	Error message displayed	Pass	Correct validation
UA-005	Password Recovery	1. Click on 'Forgot Password'2. Enter registered email 3. Submit request	Password reset link sent	Reset link sent successfully	Pass	Email functionality

4.2.2 Package Management

Test Case ID	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status	Remarks
PM-001	Add New Package	1. Login as admin 2. Navigate to Package management 3.Add Package details 4. Click Save	Package added successfully	Package added successfully	Pass	Working properly
PM-002	Edit Package	1. Login as admin 2. Navigate to Package list 3. Edit Package details 4. Click Save	Package updated successfully	Package updated successfully	Pass	Updated correctly
PM-003	Delete Package	1. Login as admin 2. Navigate to Package list 3. Click Delete	Package removed from list	Package removed successfully	Pass	Deleted successfully

4.2.3 Booking Management

Test Case ID	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status	Remarks
OP-001	Checkout Process	1. Login as customer 2. Go to cart 3. Click Checkout 4. Enter shipping details 5. Confirm Booking	Booking placed successfully	Booking placed successfully	Pass	Working properly
OP-002	Checkout Process	1. Login as customer 2. Go to cart 3. Click Checkout 4.Enter shipping details 5. Confirm Booking	Booking placed successfully	Booking placed successfully	Pass	Process repeated
OP-003	Booking Status Update	1. Login as admin 2. Navigate to Bookings 3.Update Booking status	Status updated successfully	Status updated successfully	Pass	Updated correctly

4.2.4 Admin Functionalities

Test Case ID	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status	Remarks
AD-001	Manage Users	1. Login as admin 2.Navigate to user management 3.Add/Edit/Delete user	User details managed successfully	User details managed successfully	Pass	Working properly
AD-002	View Reports	1. Login as admin 2. Go to reports section 3. View sales reports	Reports displayed successfully	Reports displayed successfully	Pass	Display working

4.2.5 Security Testing

Test Case ID	Test Scenario	Steps to Execute	Expected Result	Actual Result	Status	Remarks
SEC-001	SQL Injection	1. Enter SQL queries in input fields 2. Submit the form	System prevents SQL injection	SQL injection prevented	Pass	Secure against attack
SEC-002	Cross-Site Scripting (XSS)	1. Enter script in input fields 2. Submit the form	System sanitizes input	Input sanitized successfully	Pass	Input validation working

4.3 System Implementation

System implementation is the process of converting the system design into a working application. In the Smart Location-Based Hospital Finder System, the implementation is done using PHP as the front-end programming language and MySQL as the backend database. The application is developed and executed using the XAMPP server environment.

The implementation of a PHP–MySQL application involves connecting PHP scripts with the MySQL database to perform operations such as adding, viewing, updating, and deleting records (CRUD operations). First, a MySQL database is created and tables are defined to store data related to patients, doctors, admin, and appointments.

A configuration file called config.php is used to store the database connection details such as hostname, username, password, and database name. Using this file, PHP establishes a connection to the MySQL database through MySQLi functions.

After the connection is established, SQL queries are executed to manage and retrieve the data from the database. The user interface of the system is developed using HTML, JavaScript, and CSS, which helps users interact with the system easily. Through this implementation, the system allows users to search for hospitals, manage data, and access healthcare information efficiently.

4.4 System Maintenance

The System Maintenance Management System is a web-based application developed using PHP and MySQL to manage and automate computer system maintenance activities within an organization. The main objective of this system is to handle hardware problems, software issues, and routine system servicing in places such as offices, colleges, and IT departments.

The system provides a centralized platform where users can register complaints related to system problems. After a complaint is submitted, the administrator can review the request and assign the task to a technician. The technician is responsible for checking the system issue and updating the repair or maintenance status in the system. This helps the organization track all maintenance activities in an organized way.

The system also supports secure login and role-based access for different users such as admin, technicians, and normal users. The admin can manage user accounts, system details, maintenance schedules, and generate reports. The technicians can view assigned tasks and update repair details, while users can track the status of their complaints and view maintenance history.

By storing all information in a database, the system reduces manual record keeping and minimizes errors. It improves efficiency, transparency, and response time in handling maintenance issues. This system is useful for both academic purposes and real-time implementation in organizations that require proper IT maintenance management.

5.CONCLUSION

Hence we have achieved smart health management system using data mining. The whole systems activities are divided into three major parts like patients, doctors, and admin. Each one has their own role to perform and system respond accordingly. Several agents have been created using web services and inter agent communication is done. PhpMyadmin is used for storing data of patients, doctors, and admin.For implementing the system PHP and HTML programming is used.

The system comprise of following features.

- Management of Doctors
- Management of Patients
- Management of Schedules of Doctor
- Management of Patients Appointments
- Management of Patient – Doctor Dialogs
- Searching Information

6.SCOPE FOR FUTURE ENHANCEMENT

The Smart Location-Based Hospital Finder System helps users easily find nearby hospitals and doctors through an online platform. Patients can search for hospitals and doctors based on location, specialization, and availability. The system allows users to register, log in, and quickly access hospital and doctor information.

It also enables users to view doctor details, hospital facilities, and available services, helping them choose the right healthcare provider. The system assists doctors and administrators in managing patient details, appointments, and hospital data efficiently. All information is stored in a centralized database, which improves data accessibility, reduces manual work, and provides faster responses to users.

This system can be used in hospitals, clinics, and healthcare organizations to improve healthcare service management and make it easier for patients to find medical assistance quickly.

Future Enhancement

In the future, the system can be enhanced by adding more advanced features to improve its functionality and user experience. The system can include GPS-based location services to automatically detect nearby hospitals. Online appointment booking and payment facilities can also be added. Another enhancement could be video consultation with doctors for remote patients. The system can also be integrated with mobile applications to make it more accessible for users. Additional features like emergency ambulance services, online medical reports, and health record management can also be included to make the system more effective and useful.

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- ✓ www.tutorialpoint.com
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8.APENDIX

A. SCREEN LAYOUT

Figure A1 : Patient Login Page

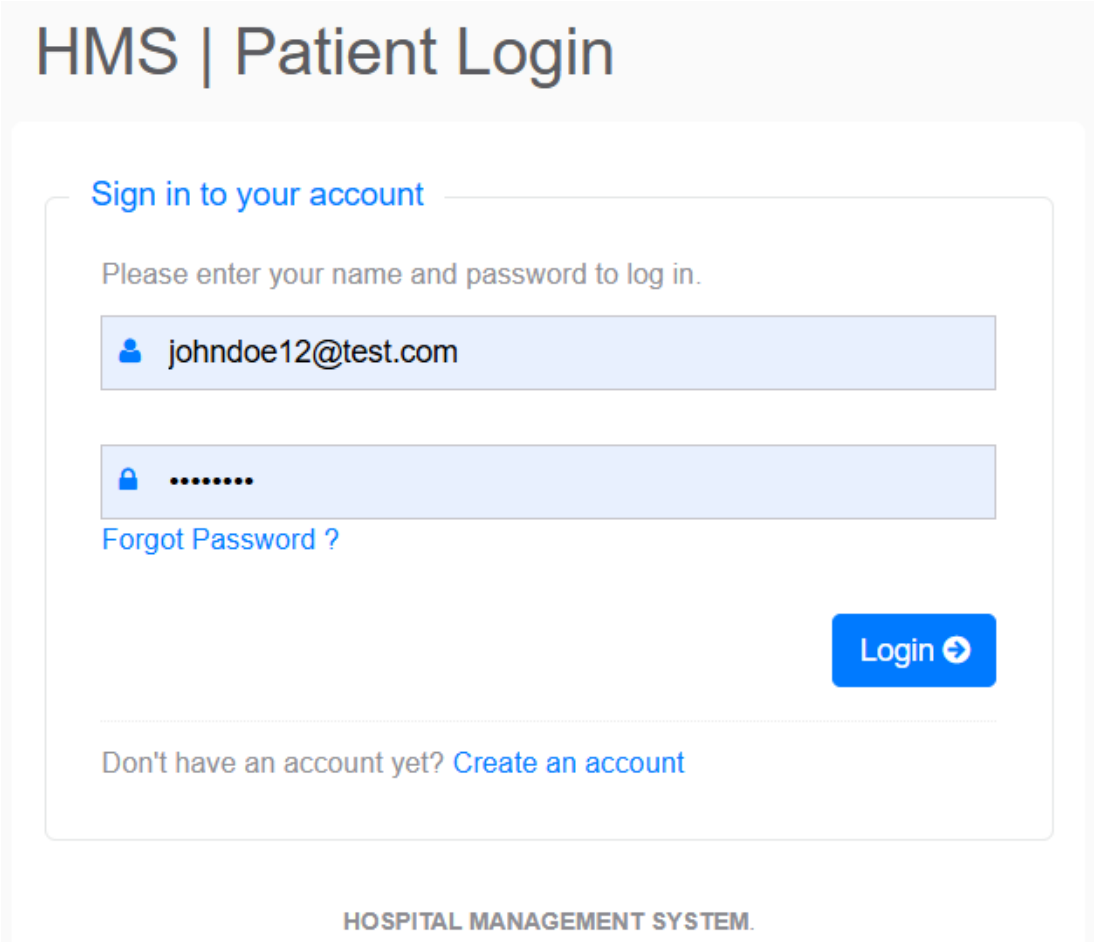


Figure A2 : Patient Registration Page

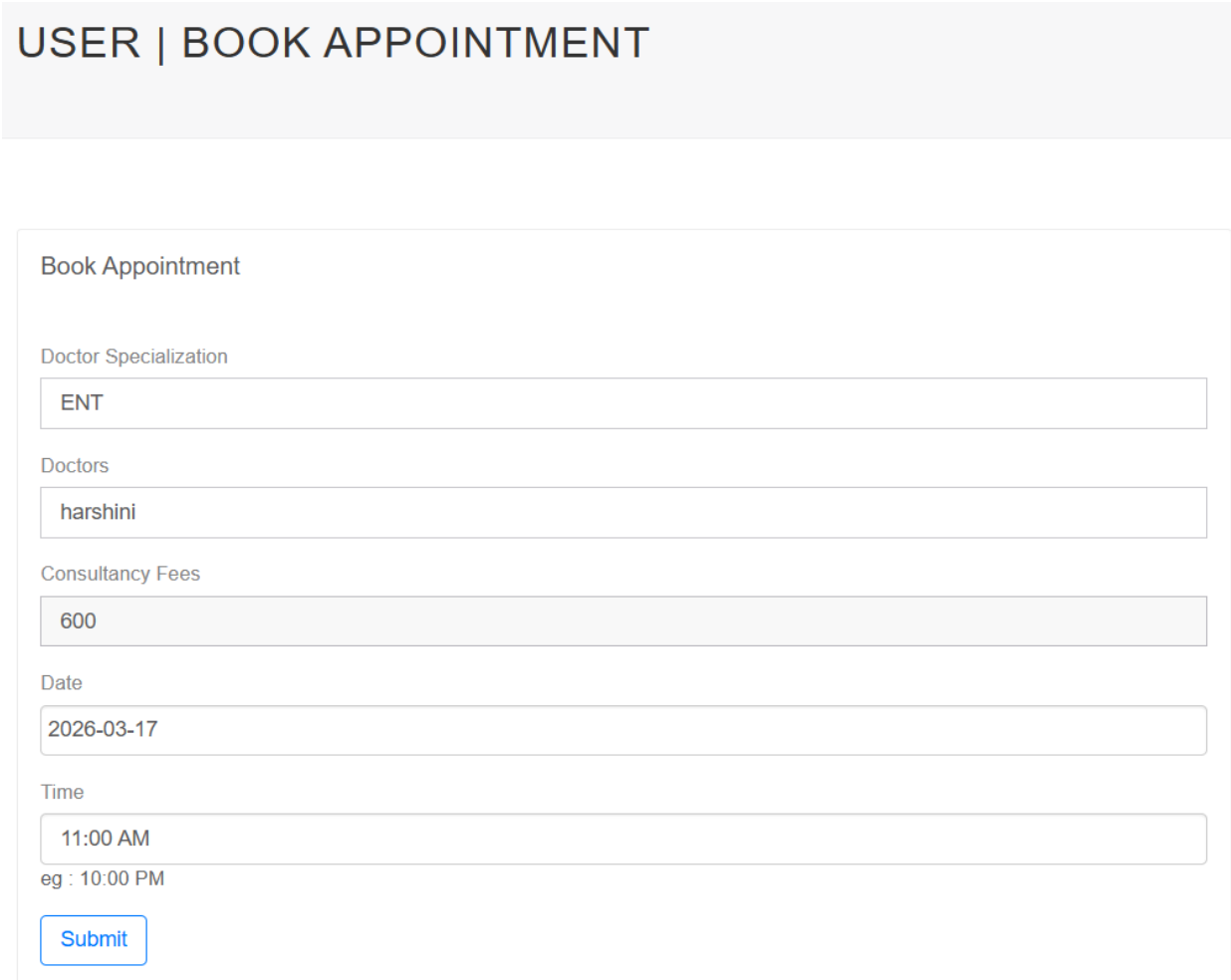


Figure A3 : Doctor Login Page

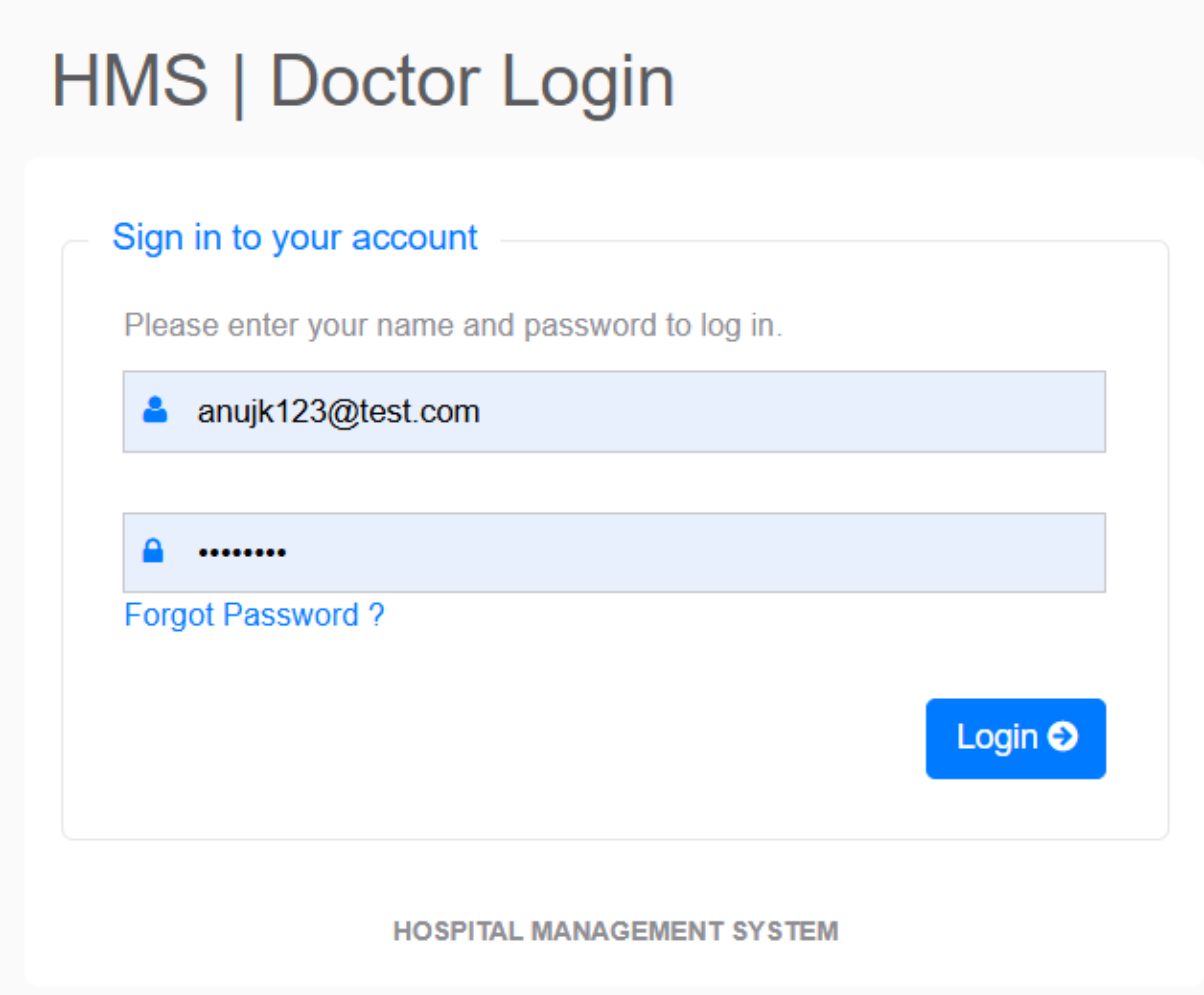


Figure A4 : Doctor Dashboard

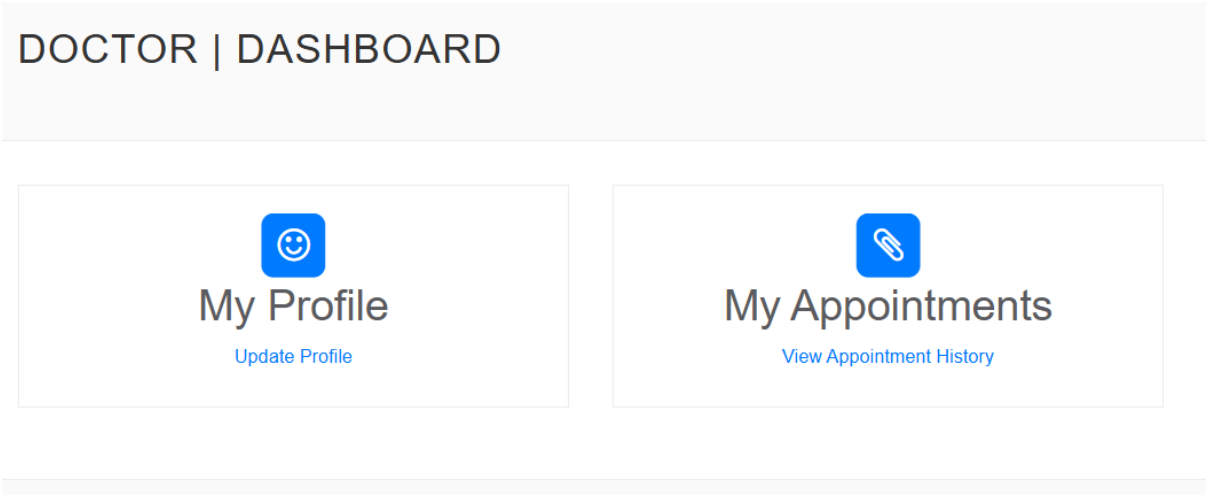
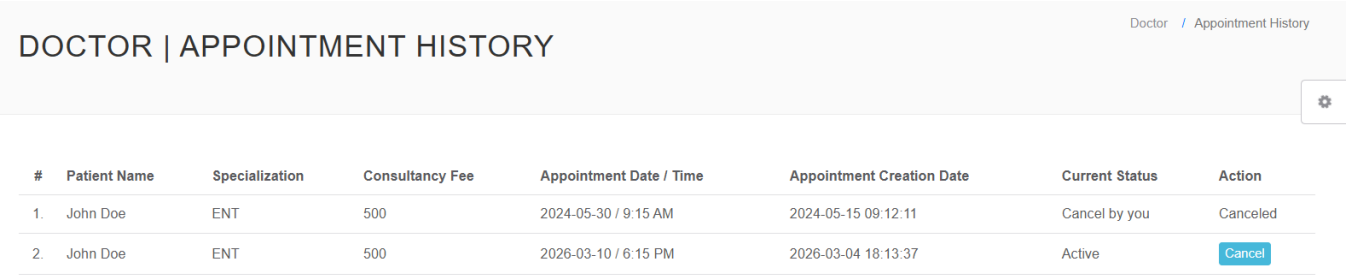


Figure A5 : DOCTOR APPOINTMENT HISTORY



#	Patient Name	Specialization	Consultancy Fee	Appointment Date / Time	Appointment Creation Date	Current Status	Action
1.	John Doe	ENT	500	2024-05-30 / 9:15 AM	2024-05-15 09:12:11	Cancel by you	Canceled
2.	John Doe	ENT	500	2026-03-10 / 6:15 PM	2026-03-04 18:13:37	Active	Cancel

Figure A6 : Add Patient

PATIENT | ADD PATIENT

Add Patient

Patient Name

Santhiya

Patient Contact no

6369772626

Patient Email

santhiyamurali70@gmail.com

Email available for Registration .

Gender

Female Male

Patient Address

No,8 p.k street,palladam,pollachi.

Patient Age

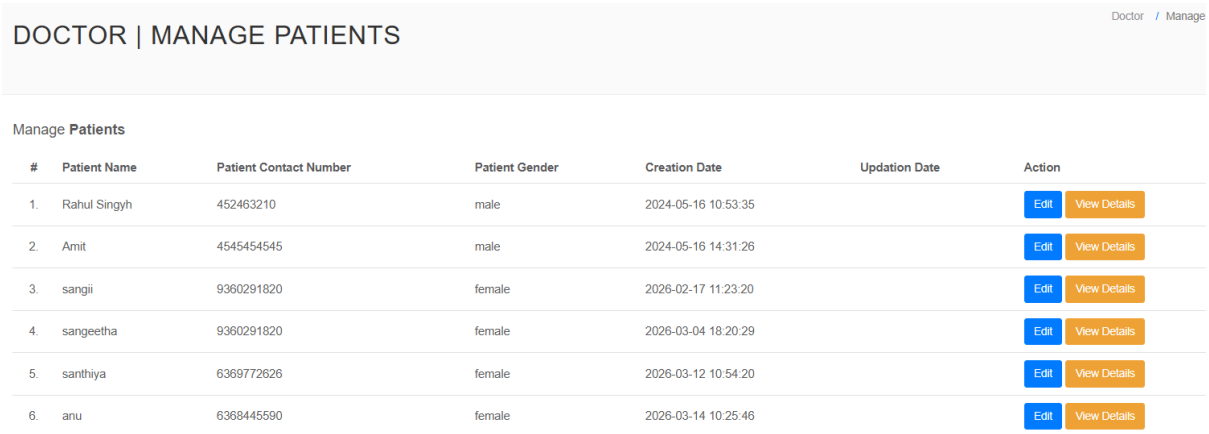
20

Medical History

fever and cold

Add

Figure A7 : Manage Patient



DOCTOR | MANAGE PATIENTS

Doctor / Manage

Manage **Patients**

#	Patient Name	Patient Contact Number	Patient Gender	Creation Date	Updation Date	Action
1.	Rahul Singyh	452463210	male	2024-05-16 10:53:35		Edit View Details
2.	Amit	4545454545	male	2024-05-16 14:31:26		Edit View Details
3.	sangii	9360291820	female	2026-02-17 11:23:20		Edit View Details
4.	sangeetha	9360291820	female	2026-03-04 18:20:29		Edit View Details
5.	santhiya	6369772626	female	2026-03-12 10:54:20		Edit View Details
6.	anu	6368445590	female	2026-03-14 10:25:46		Edit View Details

Figure A8 : Admin Login

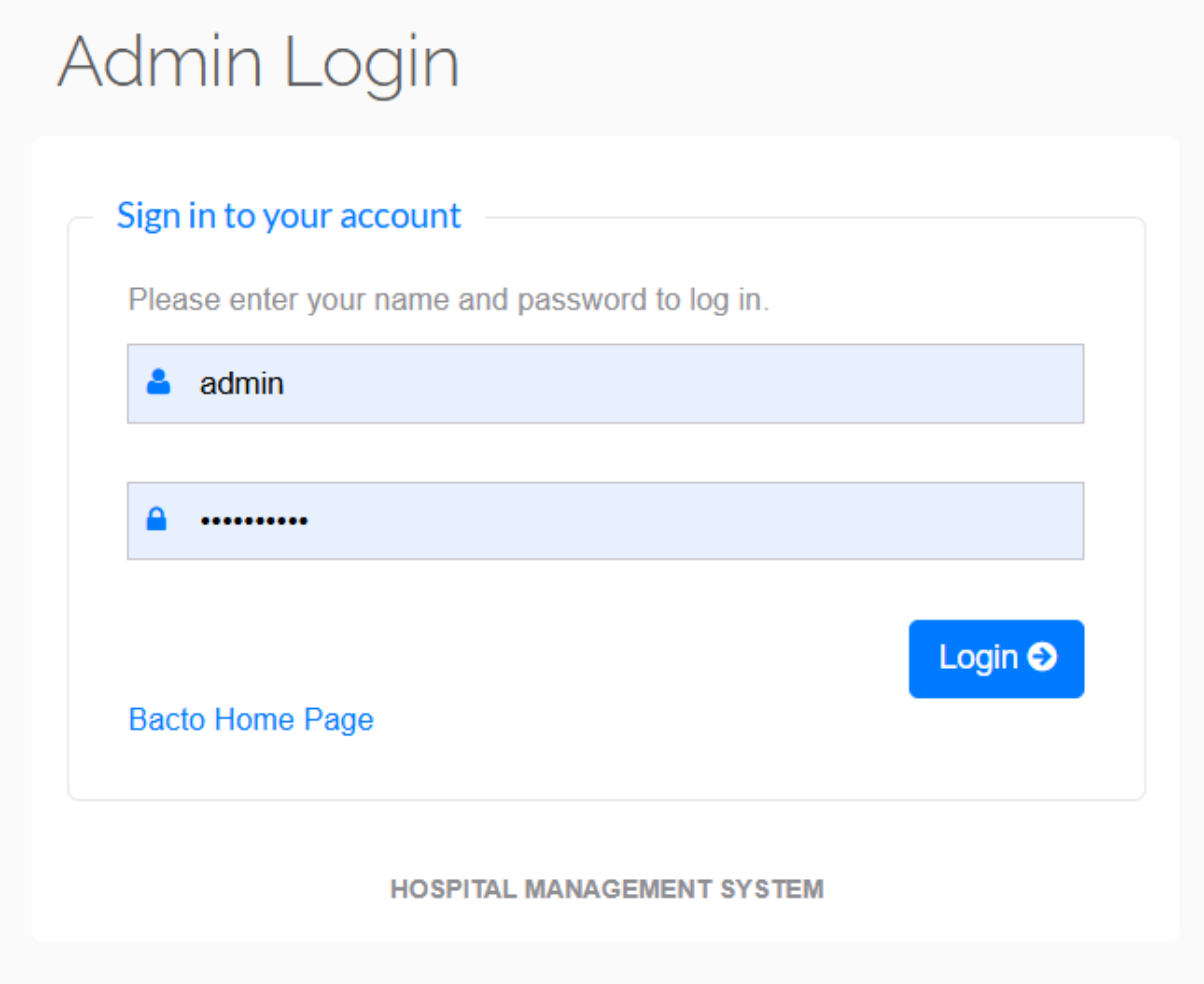


Figure A9 : Admin Dashboard

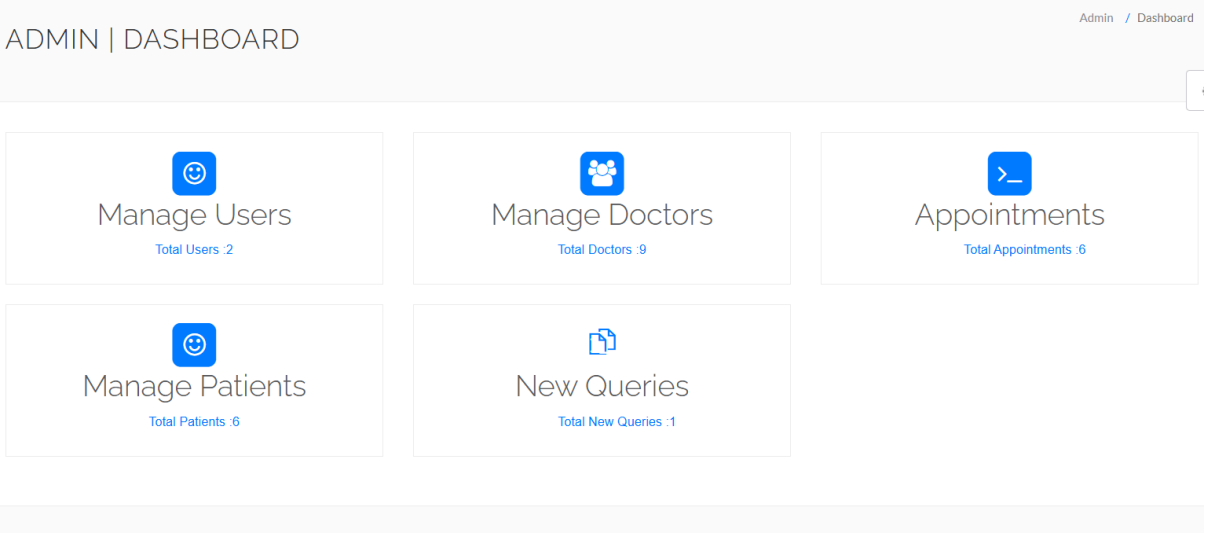


Figure A10 : Manage Doctor

ADMIN | MANAGE DOCTORS

Admin / Manage Doctors

Manage **Doctors**

#	Specialization	Doctor Name	Creation Date	Action
1.	ENT	Anuj kumar	2024-04-10 23:46:52	
2.	Endocrinologists	Charu Dua	2024-04-11 06:36:41	
3.	Pediatrics	Priyanka Sinha	2024-05-16 14:42:23	
4.	Orthopedics	Vipin Tayagi	2024-05-16 14:43:11	
5.	Internal Medicine	Dr Romil	2024-05-16 14:44:11	
6.	Obstetrics and Gynecology	Bhavya rathore	2024-05-16 14:45:18	
7.	Neurologists	santhiya	2026-02-17 11:15:09	
8.	dermatology	sangeetha	2026-03-04 17:45:13	
9.	ENT	harshini	2026-03-12 10:47:54	

Figure A11 : Doctor Specialization

Doctor Specialization

Doctor Specialization

dermatology

Submit

Manage **Docter Specialization**

#	Specialization	Creation Date	Updation Date	Action
1.	Orthopedics	2024-04-09 23:39:46	2024-05-14 14:56:47	
2.	Internal Medicine	2024-04-09 23:39:46	2024-05-14 14:56:56	
3.	Obstetrics and Gynecology	2024-04-09 23:39:46	2024-05-14 14:56:56	
4.	Dermatology	2024-04-09 23:39:46	2024-05-14 14:56:56	
5.	Pediatrics	2024-04-09 23:39:46	2024-05-14 14:56:56	
6.	Radiology	2024-04-09 23:39:46	2024-05-14 14:56:56	

Figure A12 : Add Doctor

ADMIN | ADD DOCTOR

Add Doctor

Doctor Specialization

ENT

Doctor Name

sangeetha

Doctor Clinic Address

NO,8 sk,street,kottur ,pollachi.

Doctor Consultancy Fees

600

Doctor Contact no

9360291820

Doctor Email

sangi@gmail.com

Email available for Registration .

Password

•••••••

Confirm Password

•••••••

Submit

Figure A13 : Patient Appointment History

Patients / Appointment

PATIENTS | APPOINTMENT HISTORY

#	Doctor Name	Patient Name	Specialization	Consultancy Fee	Appointment Date / Time	Appointment Creation Date	Current Status	Action
1.	Anuj kumar	John Doe	ENT	500	2024-05-30 / 9:15 AM	2024-05-15 09:12:11	Cancel by Doctor	Canceled
2.	santhiya	John Doe	Neurologists	500	2026-02-17 / 11:30 AM	2026-02-17 11:20:14	Cancel by Patient	Canceled
3.	Vipin Tayagi	John Doe	Orthopedics	1200	2026-03-24 / 6:15 PM	2026-03-04 18:09:39	Active	No Action yet
4.	Anuj kumar	John Doe	ENT	500	2026-03-10 / 6:15 PM	2026-03-04 18:13:37	Active	No Action yet
5.	harshini	John Doe	ENT	600	2026-03-18 / 11:00 AM	2026-03-12 10:51:00	Active	No Action yet
6.	Charu Dua	Amit kumar	Endocrinologists	800	2024-05-31 / 2:45 PM	2024-05-16 14:38:54	Active	No Action yet

B.SAMPLE CODING

Database Connection

```
<?php
$host = "localhost";
$user = "root";
$password = "";
$dbname = "hospital_finder";

$conn = mysqli_connect($host,$user,$password,$dbname);

if(!$conn){
   die("Database connection failed: ".mysqli_connect_error());
}
?>
```

Home Page – Enter City

```
<!DOCTYPE html>
<html>
<head>
<title>Smart Hospital Finder</title>
</head>

<body>

<h2>Smart Location-Based Hospital Finder</h2>

<form action="get_location.php" method="POST">

<label>Enter Your City:</label><br><br>
```

```
<input type="text" name="city" required>

<br><br>

<input type="submit" value="Find Hospitals">

</form>

</body>
</html>
```

Get Location Page

```
<?php
$city = $_POST['city'];
?>

<!DOCTYPE html>
<html>
<head>
<title>Select Hospital Type</title>
</head>

<body>

<h3>Location: <?php echo $city; ?></h3>

<form action="search_hospital.php" method="POST">

<input type="hidden" name="city" value="<?php echo $city; ?>">
```

```
<label>Select Hospital Type:</label><br><br>

<select name="type">

<option value="General">General</option>
<option value="Government">Government</option>
<option value="Private">Private</option>
<option value="Emergency">Emergency</option>

</select>

<br><br>

<input type="submit" value="Search Hospitals">

</form>

</body>
</html>
```

Search Hospital Page

```
<?php
include "db.php";

$city = $_POST['city'];
$type = $_POST['type'];

$query="SELECT * FROM hospitals WHERE city='$city' AND type='$type'";
$result=mysqli_query($conn,$query);
?>
```

```
<!DOCTYPE html>
<html>
<head>
<title>Hospital Results</title>
</head>

<body>

<h2>Hospitals in <?php echo $city; ?> (<?php echo $type; ?>)</h2>

<?php

if(mysqli_num_rows($result)>0){

echo "<table border='1'>
<tr>
<th>Hospital Name</th>
<th>Address</th>
<th>Contact</th>
</tr>";

while($row=mysqli_fetch_assoc($result)){

echo "<tr>
<td>".$row['name']."</td>
<td>".$row['address']."</td>
<td>".$row['phone']."</td>
</tr>";
```

```
}

echo "</table>";

}

else{

echo "No hospitals found!";

}

?>

<br>
<a href="index.php">Search Again</a>

</body>
</html>
```

View All Hospitals

```
<?php
include "db.php";

$query="SELECT * FROM hospitals";
$result=mysqli_query($conn,$query);
?>
```

```
<!DOCTYPE html>
<html>
<head>
<title>Hospital List</title>
</head>

<body>

<h2>All Registered Hospitals</h2>

<table border="1">

<tr>
<th>Name</th>
<th>City</th>
<th>Type</th>
<th>Phone</th>
</tr>

<?php

while($row=mysqli_fetch_assoc($result)){

echo "<tr>
<td>".$row['name']."</td>
<td>".$row['city']."</td>
<td>".$row['type']."</td>
<td>".$row['phone']."</td>
</tr>";
```

```
}

?>

</table>

</body>
</html>
```

Database Table

```
CREATE TABLE hospitals (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
city VARCHAR(50),
type VARCHAR(50),
address TEXT,
phone VARCHAR(15)
);
```